

NAIROBI JOMO KENYATTA INTL, Kenya

WMO: 637400

Lat: 1.319S

Lon: 36.928E

Elev: 5327

StdP: 12.08

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	50.4	53.1	39.2	42.9	78.8	42.5	48.9	75.4	18.0	71.2	15.8	71.2	4.5	240	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	21.1	84.6	60.8	83.2	60.7	82.1	60.8	66.3	74.7	65.6	73.9	65.0	73.4	13.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.3	110.3	67.7	63.3	106.3	66.9	62.9	104.7	66.5	34.3	75.5	33.7	74.2	33.2	72.7	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	18.6	16.7	46.1	88.6	1.8	2.7	44.8	90.5	43.8	92.1	42.8	93.6	41.5	95.5		
(5)			44.6	68.6	1.6	1.3	43.4	69.6	42.5	70.3	41.6	71.1	40.5	72.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.8	69.1	70.3	70.9	69.4	67.5	65.2	63.8	64.7	67.2	69.1	68.2	68.4
	DBStd	3.00	1.96	2.15	2.15	1.73	1.84	2.18	2.46	2.33	2.46	2.13	1.94	1.79
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	126	1	0	0	0	3	22	52	34	9	2	2	1
	CDD50	6499	591	568	648	582	544	457	428	457	515	593	547	569
	CDD65	1151	127	148	183	132	82	29	16	26	74	130	99	105
	CDH74	6593	933	1203	1269	497	205	114	131	190	540	768	309	434
	CDH80	827	112	241	263	33	4	2	4	6	50	86	10	16
Wind	WSAvg	7.8	9.2	9.5	9.3	7.5	6.2	5.9	6.0	6.2	7.0	8.5	8.8	9.3
Precipitation	PrecAvg	29.0	2.3	1.9	3.0	5.4	4.1	1.2	0.5	0.6	0.7	1.7	4.8	3.6
	PrecMax	50.8	14.4	11.4	8.0	13.7	14.6	8.9	2.5	2.5	3.1	6.4	19.1	13.0
	PrecMin	12.9	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	1.0	0.2
	PrecStd	8.9	2.8	2.2	2.2	3.1	3.3	1.5	0.6	0.6	0.7	1.5	3.4	3.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.5	86.4	87.0	83.1	80.2	79.2	80.4	80.7	83.9	84.2	81.0	81.9
		MCWB	59.9	60.1	60.9	62.1	62.9	62.3	61.4	60.9	60.4	61.4	62.1	60.8
	2%	DB	82.6	84.5	84.6	80.6	77.6	77.0	77.3	78.4	81.0	82.4	79.1	80.2
		MCWB	60.0	60.2	61.3	63.2	63.1	61.8	60.4	60.3	60.3	60.6	62.0	61.4
	5%	DB	80.9	82.8	82.8	78.8	76.5	75.1	75.1	76.0	79.2	80.6	77.3	78.5
		MCWB	60.7	60.3	61.4	63.2	63.1	61.6	60.2	60.0	60.3	61.0	62.4	61.6
	10%	DB	79.1	81.0	80.9	77.0	74.9	73.3	72.9	73.6	77.3	78.7	75.5	76.8
		MCWB	60.9	60.8	62.0	63.4	63.0	61.0	59.5	59.6	60.0	60.7	62.5	62.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.1	66.6	66.6	67.2	66.6	65.4	63.5	63.7	64.3	65.2	66.0	66.4
		MCDB	75.2	76.2	76.4	75.9	73.8	73.8	74.7	75.0	77.0	75.8	73.2	74.2
	2%	WB	64.8	65.3	65.5	66.2	65.6	63.9	62.1	62.1	62.7	64.0	64.8	65.1
		MCDB	73.9	74.7	75.3	74.0	72.7	72.5	73.1	73.1	75.4	74.1	72.4	72.9
	5%	WB	64.0	64.3	64.8	65.4	64.7	62.8	61.2	61.2	61.9	63.1	64.3	64.4
		MCDB	73.0	73.9	74.4	73.1	72.1	71.1	71.5	72.0	74.1	72.5	71.8	72.2
	10%	WB	63.1	63.4	64.1	64.7	64.1	61.9	60.2	60.2	61.1	62.4	63.7	63.7
		MCDB	72.2	73.2	73.5	72.3	71.2	69.9	69.7	70.4	72.8	71.5	70.7	71.0
Mean Daily Temperature Range	5% DB	MDBR	21.8	23.3	21.1	17.2	16.2	17.3	18.6	18.8	21.7	20.5	16.0	17.8
		MCDBR	24.9	26.1	24.3	19.5	18.6	20.7	23.6	23.9	25.2	23.6	18.6	20.6
		MCWBR	7.5	7.8	6.6	5.6	6.4	8.2	9.4	9.0	8.0	6.4	5.1	6.0
	5% WB	MCDBR	18.9	20.7	19.4	16.4	15.5	18.0	20.2	20.4	22.2	19.5	15.5	16.1
		MCWBR	6.4	6.9	6.0	5.7	6.1	7.6	8.5	8.1	7.9	5.8	5.2	5.5
Clear-Sky Solar Irradiance	taub	0.371	0.399	0.381	0.329	0.327	0.365	0.379	0.364	0.345	0.356	0.352	0.364	
	taud	2.437	2.337	2.417	2.629	2.625	2.467	2.389	2.422	2.480	2.456	2.508	2.476	
	Ebn at Noon	305	296	298	308	301	285	282	293	305	305	308	305	
	Edh at Noon	38	43	39	30	30	34	37	37	36	37	35	36	
All-Sky Solar Radiation	RadAvg	2052	2144	2079	1840	1681	1484	1461	1558	1884	1926	1787	1899	
	RadStd	133	107	137	95	64	104	138	115	109	94	95	115	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.85	+1.24	N/A	+1.03	+0.70	+0.29	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.85	+1.24	N/A	+1.03	+0.70	+0.29	N/A	N/A	N/A

Nomenclature: See separate page